

Ugence Labs — One-Page Brief

Generic platform overview. No recipient-specific framing; safe to send as-is.

The thesis

Modern enterprise AI has excellent parts and a missing middle. Foundation models reason. Orchestrators wire. Clouds host. None of them determine, deterministically and before commit, whether the exact action an autonomous agent proposes is allowed to happen — and none of them hold the credential that would make that decision enforceable.

That gap is what stops enterprises from deploying autonomous agents into anything consequential: payments, production databases, infrastructure changes, patient records, physical actuation.

Ugence Labs builds the governed-execution platform that closes it. Three architectural layers, nine platform components, one loop.

The architecture in three layers

- **Specialized AI systems** — runtimes that reason, steer, and execute: a digital-agent runtime and a physical-autonomy runtime, sharing a long-context reasoning substrate.
- **AI control plane** — products that externally govern every action the runtimes propose: context admissibility, deterministic per-action authorization, and clearance against live operational state.
- **AI infrastructure** — memory and scaling substrates that make the whole thing affordable and refuse to scale into a failure.

The **loop** is what is differentiated, not any single module: runtime *proposes* → control plane *governs* → infrastructure *runs* → world *responds* → runtime *learns*. Every hand-off is a clean boundary owned by exactly one product, which is what makes the platform auditable end-to-end.

Current engineering state — built, integrated, and internally validated

Advanced engineering across the platform, at multiple stages of readiness. Every claim below is traceable to internal validation; none of it is externally deployed.

- **Agentic Framework v1.10** — governed digital-execution runtime. 1,550+ internal tests. The **cancel → budget → approve → execute** invariant is a tested runtime contract. Two internal pilots complete. *Pilot-ready*.
- **ActionGate reference implementation** — deterministic per-action authorization. Kubernetes control-plane integration over authenticated REST/mTLS; MCP protocol integration; 24/24 conformance vectors; 12/12 injected attacks detected at real detection points. Independent architectural validation returned **SUPPORTED WITH LIMITATIONS**. Ed25519 signing, durable replay/commit stores, signed custody-separated audit ledger, internal red-team suite. *Pilot-oriented reference implementation; TRL 4 overall, isolated hardened subsystem ~TRL 5*.
- **Context Minimization** — extractive, fail-closed context reduction using ActionGate as its deterministic authorization oracle. Frozen cross-model benchmark on three real open-weights models: 32-50% token reduction at 100% authorization-decision preservation. *Cross-model validated research prototype; recommendation LIMITED_GO*.
- **KV Pro** — compression engine shipped via vLLM as a registered KV-cache backend. Eviction engine (CTM+/PCAM) software-production-ready; +50% concurrent users, -29% p99 vs LRU on real Mistral-7B and Llama-3.1-8B KV data. *Shipped (compression); production-ready (eviction)*.
- **Neural Cloud Scaling Controller** — decision-quality layer above every autoscaler. Zero SLO regressions across 19 adversarial scenarios in simulated validation. Shadow mode, recommend mode, and a live-shadow harness are built. *Software-production-ready; first real-cluster pilot pending*.
- **Autonomous Robotics — BCVF** — predictor-trust runtime for safety-critical autonomy. 1,117 tests; 0% FPR / 0% FNR on a 1,560-cell certification-grade characterization grid (Wilson 95% CI floor 0.90). SOTIF / ISO 26262 traceability matrix is a shipping deliverable. *Research prototype; first real-sensor pilot pending*.
- **LLM Steering Controller (CSR) + CG LLM research architecture** — model-agnostic frame-control and answer-audit layer, validated on one open model; the deeper CG LLM interpretable-generation architecture (partial implementation) sits behind it. *Mixed — CSR pilot-ready; CG LLM research-stage*.
- **Hybrid LLM** — long-range attention research; serial-fusion architecture. 100% needle-in-haystack accuracy at 2K and 10K token distances on a controlled retrieval task. Training stack built end-to-end. *Research-stage*.
- **PSE — Phoneme Symbolic Engine** — deterministic naming and verbal-identity engine, built and byte-identity-tested. Commercial surfaces (studio, observation platform) are the build ahead. *Standalone vertical*.

Aggregate: 4,700+ tests across the platform. **Pre-revenue and pre-external-deployment;** actively onboarding first design partners.

What's next

First external design-partner deployments on Kubernetes-based infrastructure (ActionGate + Context Minimization together as the control-plane pilot); first regulated-enterprise Agentic Framework pilot; first real-cluster Cloud Scaling deployment; robotics real-sensor pilot. Third-party architectural reviews and independent audits follow.

What we're looking for

1. **Signal on the positioning** — the "governed execution as a distinct platform layer" thesis is one we want to pressure-test with people who have evaluated enterprise-AI platforms as buyers.
2. **Design partners** — organizations deploying tool-calling or autonomous agents into consequential workflows; preferably Kubernetes / MCP-based or in a regulated environment.

About

Ugence Labs. Based at T-Hub, Hyderabad. Solo founder with a small team; deep engineering across the stack is the moat.

Deeper material available on request: a Platform Architecture Overview (13 pp), a customer-facing brochure (18 pp), and a pitchbook (72 pp). See `UGENCE_PACKAGE_IP_INDEX.md` for the module-by-module engineering inventory.

Ugence — Package IP Index

Every independently packaged module under `packages/`, one or two lines each. Companion to `UGENCE_PLATFORM_ONE_PAGER.md`. Descriptions are taken from each package's own README; they describe what is **built and packaged**, not what is externally deployed. The platform is pre-revenue and pre-external-deployment.

Governance kernel — the authority layer

Module	What it is
<code>ugence-risk-authority</code>	Turns an approved governance decision into cryptographically bound, scoped, time-bound, revocable runtime authority — and enforces it at the exact point of action. The sole machine-execution authority artifact.
<code>ugence-decision-authority</code>	Domain-neutral governance kernel deciding when an AI recommendation may become a binding business decision: decision cases, authorization, execution, reconciliation.
<code>ugence-policy-authority</code>	The one platform-wide policy authority: issues, signs, registers, resolves, verifies and revokes policy versions for any registered policy family.
<code>ugence-governance-contracts</code>	Canonical neutral contract layer every capability depends on, so capabilities never depend on each other. Leaf package, no authority of its own.
<code>ugence-governance-provider-framework</code>	Registration, resolution, invocation, observation and conformance testing of governance providers. Owns no governance authority.
<code>ugence-trusted-evidence-authority</code>	Trusted-evidence contracts plus the verification authority: trust anchors, key trust and revocation, signing, independent verification.
<code>ugence-benchmark-registry</code>	Frozen benchmark identity and lifecycle contracts — the shared, platform-wide benchmark definition layer.
<code>ugence-benchmark-registry-authority</code>	Registry and exact-resolution contracts above the frozen identity layer. Resolves; computes nothing.
<code>ugence-uvi-policy-contracts</code>	Immutable contract shapes for Ugence Value Intelligence governed-assessment context and policy artifacts. Schema layer only.

Governance providers — pluggable evaluators

Module	What it is
<code>ugence-actiongate-provider</code>	ActionGate — action-governance provider: given a proposed action and its authority, policy, risk, evidence and decision context, returns authorized / authorized-with-constraints / denied / indeterminate.
<code>ugence-tap-provider</code>	TAP — assertion-governance provider: given a material assertion and evidence references, returns supported / unsupported / constrained / indeterminate, component by component.

Control-plane capabilities

Module	What it is
<code>ugence-context-minimization</code>	Deterministic, extractive, fail-closed context reduction that preserves a caller-defined equivalence condition. Never generative; fails closed when equivalence cannot be established.
<code>ugence-action-clearance</code>	Decides whether an already-authorized exact action remains operationally CLEAR immediately before execution. May narrow, hold, escalate or block — never broaden.
<code>ugence-model-selection</code>	Model Authority — determines which model, if any, is <i>permitted</i> to execute the current request under policy and runtime constraints.
<code>ugence-llm-steering-controller</code>	Deterministic, provider-neutral, advisory-only LLM routing: candidate discovery, fail-closed hard constraints applied before scoring, then ranking.
<code>ugence-storygraph</code>	Advisory sequence-risk analyzer: detects when individually-acceptable actions assemble a prohibited capability, and emits evidence for a downstream authority to escalate.
<code>ugence-agent-workforce-composer</code>	Deterministic offline planning: which workflow nodes may be performed by AI agents, what capabilities those roles require, which registered agents are eligible under frozen hard constraints.
<code>ugence-agent-value-readiness</code>	Experimental, advisory readiness-determination evaluator with a fail-closed trust boundary. Not a deployment authority.
<code>ugence-governed-value</code>	Experimental post-deployment value-calculation kernel over caller-reported, unverified inputs. Cannot claim a figure is observed, attributed or verified.
<code>ugence-policy-workflow-compiler</code>	Compiles a reviewed governance policy pack into a deterministic governed-workflow artifact plus an assurance package: workflow IR, assurance manifest, test scenarios, audit schema, structural diffs, content-addressed package.

Execution runtime

Module	What it is
<code>ugence-agent-runtime</code>	Domain-neutral execution-coordination kernel for agent and workflow execution. Deliberately excludes agent planning, reasoning and memory.

Cloud scaling — advisory through controlled execution

Module	What it is
ugence-cloud-scaling-controller	Deterministic, provider-neutral, advisory-only adaptive scaling controller producing explainable recommendations from normalized observations.
ugence-cloud-scaling-operations	The controlled-execution layer: in LIVE mode, with credentials and explicit external authorization, it can patch Kubernetes deployment scale and trigger ArgoCD syncs.
ugence-cloud-scaling-risk-integration	Phase 4C — one-way, non-executing adapter projecting a capacity recommendation into the neutral Risk Authority subject-risk contract; stops at a non-executable risk decision.
ugence-cloud-scaling-producer-attestation	Phase 5B-0A — establishes <i>who produced</i> a recommendation. A verified attestation grants nothing.
ugence-cloud-scaling-policy-authenticity	Phase 5B-0B — establishes that one exact policy version was authentically issued and valid at an injected instant. A verified proof grants nothing.
ugence-cloud-scaling-authorization-contracts	Phase 5A — turns a reconciled projection, risk binding, attestation evidence and policy binding into an explicitly non-authoritative capacity-authorization <i>candidate</i> .

Risk Authority runtime integrations (RA-4.5 → RA-8)

Module	What it is
ugence-risk-authority-runtime	RA-4.5 — fail-closed composition of the machine-authority owner with Decision Authority and ActionGate into a single execution-eligibility decision.
ugence-risk-authority-evidence-runtime	RA-5 — trusted evidence admission and control assurance, closing the gap where a caller could self-assert a control status.
ugence-risk-authority-status-runtime	RA-6 — post-issuance authority lifecycle: revocation and epoch propagation around the single signed authorization envelope.
ugence-risk-authority-runtime-assurance	RA-7 — observes runtime trajectories through the neutral event seam and emits authority-reassessment signals on material deviation. Observes; does not own consequences.
ugence-risk-authority-execution-assurance	RA-8 — post-execution reconciliation: did the actual effect match what was authorized, and should the discrepancy reassess future machine authority?
ugence-context-minimization-token-accounting-runtime	CM-TA1 — narrow one-way integration wiring Context Minimization token accounting to Agent Runtime provider-attempt telemetry and budgets.

Governance verticals

Module	What it is
ugence-ai-hiring	AI-assisted hiring governance product: audited workflow state machine, deterministic evidence normalization, decision cases, and the AI/human separation enforced in types, services, persistence and API permissions.
ugence-procurement	Governed purchase approvals and authorized supplier actions on the Decision Authority kernel, with the same separation enforced in code rather than documentation.

The pattern across all of it: each module owns exactly one decision, states plainly what it does *not* grant, and hands off across a typed boundary. That is what makes the loop — propose → govern → run → respond → learn — auditable end-to-end.